Python Mini Project (EDA Based)

Case Study: Airbnb New York City Listings Analysis

Background

Short-term rental platforms such as Airbnb have significantly changed the housing and tourism landscape of large cities. City authorities, hosts, and travelers rely on data to understand pricing trends, availability, and neighborhood-level patterns.

In this project, you will analyze a real dataset from a Kaggle competition containing Airbnb listings from New York City. Your task is to perform Exploratory Data Analysis (EDA) to identify meaningful trends and insights.

Learning Objectives

- Explore and clean a real-world dataset with mixed data types
- Apply univariate, bivariate, and multivariate EDA techniques
- Understand pricing and availability patterns using data
- Present insights using appropriate visualizations
- Develop analytical thinking through data interpretation

Dataset Information

Dataset Name: Airbnb New York City Listings (Kaggle Dataset)

The dataset contains detailed information about Airbnb listings, including location, room type, price, availability, number of reviews, and host-related attributes.

Project Instructions

Phase 1: Data Understanding

- Load the dataset and inspect its structure
- Review column names, data types, and dataset size
- Display sample records to understand the data context

Phase 2: Data Quality Assessment

- Identify missing values and duplicate records
- Detect unrealistic values such as zero or extremely high prices
- Summarize key data quality issues

Phase 3: Data Cleaning

- Handle missing values using appropriate strategies
- Remove or treat outliers in pricing and availability
- Document and justify all cleaning steps

Phase 4: Feature Preparation

- Convert categorical variables where necessary

- Create derived features if useful for analysis

Phase 5: Univariate Analysis

- Analyze the distribution of prices, room types, and availability
- Use suitable plots to describe individual variables

Phase 6: Bivariate Analysis

- Analyze price variations across neighborhoods and room types
- Study relationships between reviews, availability, and pricing

Phase 7: Multivariate Analysis

- Explore interactions between location, room type, and price
- Use grouping and pivot tables to support analysis

Phase 8: Insight Development

- Write at least 10 clear, data-supported insights
- Ensure each insight is linked to analysis or visualization

Phase 9: Final Conclusion

- Summarize major findings
- Discuss implications for hosts, guests, or city planners
- Mention limitations of the dataset

Submission Guidelines

- Submit a Jupyter Notebook (.ipynb) file
- Use clear section headings and markdown explanations
- All visualizations must include titles and axis labels
- Do not apply machine learning techniques

Evaluation Criteria

- Data cleaning and preprocessing quality
- Depth of exploratory analysis
- Visualization clarity and relevance
- Insight quality and reasoning
- Overall organization and presentation